Nikola Janjušević

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Research Experience

NYU Langone Radiology, Postdoc Fellow
Learning-based longitudinal and low-field MRI joint-reconstruction and denoising.
Fall 2024 - Present, New York, NY

NYU Langone Radiology, NTV Intern
Unsupervised DNN-based MRI denoising.
Summer 2023 - Summer 2024, New York, NY

Apple Video Engineering, Research Intern
"White-box" reference-guided image enhancement.
Summer 2022, Cupertino, CA

Teaching Experience

The Cooper Union, Adjunct Professor
ECE-150 Digital Logic Design, Fall 2022-2025
MA-110 Introduction to Linear Algebra, Fall 2025
New York, NY

NYU Tandon, Adjunct Professor
ECE-UY 4563 Introduction to Machine Learning.
Fall 2025, Brooklyn, NY

NYU Tandon, Teacher's Assistant
ECE-GY 6123 Image and Video Processing.
Spring 2022, Spring 2023, Brooklyn, NY

NYU Summer STEM, Senior Instructor
Introduction to Machine Learning.
Summer 2019, Brooklyn, NY

Awards and Honors

NYU Shiv Panwar Scholarship 2021-2023
Telephonics Research Fellowship 2020
NYU K-12 STEM Fellowship 2019
Radio Club of America Scholarship 2019
CU Half-Tuition Scholarship 2015-2019
CU Innovator's Merit Scholarship 2015-2019

Education

New York University
Ph.D. Electrical Engineering, GPA: 3.90/4.00
Thesis: *Noise Adaptive Dictionary Learning Based Deep Neural Networks for Imaging Inverse Problems*
Advisor: Professor Yao Wang, NYU Video Lab
Fall 2019 - Summer 2024, Brooklyn, NY

The Cooper Union
Bachelors of Engineering, Electrical Engineering
Magna Cum Laude, Minor in Computer Science
Fall 2015 - Spring 2019, New York, NY

Selected Graduate Courses:
Math-GA 20(10,20) Numerical Methods I, II
Math-GA 2012 Non-smooth and Convex Optimization
Math-GA 2012 High Performance Computing

Publications

[1] N. Janjušević, H. Chandarana, Y. Wang, and Li Feng, *LGGS: Longitudinal MRI Reconstruction via Guided Group Sparsity*, arXiv preprint, 2026.

[2] N. Janjušević, A. Khalilian-Gourtani, Y. Wang, and Li Feng, *Noisy MRI Reconstruction via MAP Estimation with an Implicit Deep-Denoiser Prior*, IEEE ICIP, 2026.

[3] N. Janjušević, J. Chen, L. Ginocchio, et al., *Self-Supervised Noise Adaptive MRI Denoising via Repetition to Repetition (Rep2Rep) Learning*, Magnetic Resonance in Medicine, 2025.

[4] N. Janjušević, A. Khalilian-Gourtani, A. Flinker, Li Feng, and Y. Wang, *GroupCDL: Interpretable Denoising and Compressed Sensing MRI via Learned Group-Sparsity and Circulant Attention*, IEEE TCI, 2025. Julia code.

[5] N. Janjušević, A. Khalilian-Gourtani, Y. Wang, and Li Feng, *Learned Primal Dual Splitting for Self-Supervised Noise-Adaptive MRI Reconstruction*, IEEE ISBI, 2025.

[6] N. Janjušević, A. Khalilian-Gourtani and Y. Wang, *CDLNet: Noise-Adaptive Convolutional Dictionary Learning Network for Blind Denoising and Demosaicing*, IEEE OJSP 2022. PyTorch code.

[7] N. Janjušević, A. Khalilian-Gourtani and Y. Wang, *Gabor is Enough: Interpretable Deep Denoising with a Gabor Synthesis Dictionary Prior*, IEEE IVMSP 2022. PyTorch code.

Conferences and Talks

14/09/26 N. Janjušević et al., *Noisy MRI Reconstruction via MAP Estimation with an Implicit Deep-Denoiser Prior*, Oral Presentation, IEEE International Conference on Image Processing (ICIP), September 13th - September 17th 2026, Tampere, Finland.

09/05/26 International Society for Magnetic Resonance in Medicine (ISMRM), May 9th - May 14th 2026, Cape Town International Convention Centre, Cape Town, South Africa.

- N. Janjušević et al., *Longitudinal MRI Reconstruction Leveraging Patient-Specific Group Sparsity from Prior Images*, Flash Presentation.
- N. Janjušević et al., *Diffusion-Style Noisy MRI Reconstruction via Stochastic MAP Estimation with an Implicit Denoiser Prior*, Digital Poster.
- R. Luo, N. Janjušević et al., *Noise-Adaptive Self-Supervised Denoising for Metal Artifact Reduction in Accelerated Cervical Spine MRI*, Oral Presentation.
- R. Luo, N. Janjušević et al., *Shuffled Repetition-to-Repetition Learning (Rep2Rep-Shuffle) for Noise-Adaptive Self-Supervised Denoising in Sodium MRI*, Digital Poster.

08/08/25 A. Khalilian-Gourtani and N. Janjušević, Directly Parameterized Neural Network Construction for Generalization and Robustness, IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR) 2025, San Jose, CA, USA.

17/07/25 N. Janjušević and A. Khalilian-Gourtani, SPS Webinar: Directly Parameterized Neural Network Construction for Generalization and Robustness in Imaging Inverse Problems, IEEE Signal Processing Society, 2025.

10/05/25 International Society for Magnetic Resonance in Medicine (ISMRM), May 10th - May 15th 2025, Hawai'i Convention Center, Honolulu, Hawai'i, USA.

- N. Janjušević et al., *Noise-Adaptive MRI Denoising Using Self-Supervised Learning with Average-to-Average (Avg2Avg) Loss*, Digital Poster.
- N. Janjušević et al., *Joint MRI Reconstruction and Denoising Using Noise-Adaptive Self-Supervised Learning*, Digital Poster.

17/04/25 N. Janjušević et al., *Learned Primal Dual Splitting for Self-Supervised Noise-Adaptive MRI Reconstruction*, Poster Presentation, IEEE International Symposium on Biomedical Imaging (ISBI), April 14th - April 17th 2025, Houston, Texas, USA.

27/05/24 L. Bojić and N. Janjušević, *Strategija Razvoja Veštačke Inteligencije Kod Nas / An Artificial Intelligence Strategy for Serbia*, Eduka Institute of Organizational Business, Belgrade, Serbia.

04/05/24 International Society for Magnetic Resonance in Medicine (ISMRM), May 4th - May 9th 2024, Suntec Singapore Convention & Exhibition Centre, Singapore.

- N. Janjušević et al., *Advanced Deep Learning Denoising for Accelerated 0.55T Prostate MRI*, Digital Poster + Power-Pitch Presentation.
- N. Janjušević et al., *Self-Supervised Noise-Adaptive Convolutional Dictionary Learning for Low-Field MRI Denoising*, Digital Poster.

21/02/24 N. Janjušević et al., *Learning Deep Denoisers for Low-Field MRI with Noisy Data*, Research Seminar, The Center for Biomedical Imaging (CBI) and Center for Advanced Imaging Innovation and Research (CAI^{2}R), Translational Research Building, NYU Langone Health, New York, NY, USA.

18/10/23 N. Janjušević et al., *Self-Supervised Low-Field MRI Denoising via Spatial Noise Adaptive CDL-Net*, Power-Pitch Presentation, i2i Workshop, CAI^{2}R, October 18th - October 19th 2023, NYU Langone, NY, USA.

26/06/22 N. Janjušević et al., *Gabor is Enough: Interpretable Deep Denoising with a Gabor Synthesis Dictionary Prior*, Research Presentation, IEEE 14th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP), June 26th - June 29th 2022, Nafplio, Greece.

Last updated: October 2026